

3.4 Paper 2 sigma/pi and bond energy homework — answers

Section A — Multiple choice

A1 = B

A2 = A

A3 = C

A4 = C

A5 = D

A6 = B

A7 = C

A8 = B

Section B — Mark scheme clips

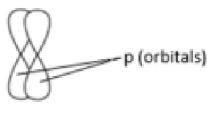
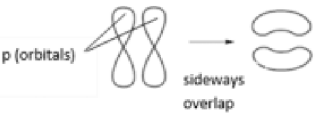
B1(a)(ii) [4]

1(a)(ii)		σ bond	π bond	4
	no of bonds	1	2	
	how orbitals overlap	direct (orbital overlap)	sideways (overlap of adjacent p orbitals)	

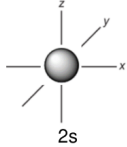
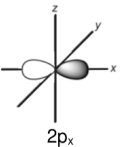
B2(b)(i) [1]

5(b)(i)	sp^2	1
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B3(b)(ii) [2]

5(b)(ii)	M1 σ bond – diagram showing how σ bond is made from direct overlap of orbitals i.e.  OR  M2 π bond - diagram showing how π bond is made from sideways overlap of 2 $\times$ p-orbitals i.e.  OR  OR  sideways overlap	2
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B4(c)(iii) [2]

1(c)(iii)	 	2
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B5(b) [1]

2(b)	CO / it is a polar molecule / it has a (permanent) dipole (but N_2 is non-polar)	1
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B6(d)(ii) [2]

Question	Answer	Marks
3(d)(ii)	M1 HCl is more stable AND (because) Cl atom is smaller / less distance between electron pair and nucleus / shorter (H-X) bond length M2 (In HCl) stronger attraction between nucleus and shared pair of electrons OR (In HCl) (fewer shells of electrons so) less shielding OR (In HCl) greater orbital overlap so stronger interaction	2